

Anqi (Angie) Zhu, PhD

SUMMARY

- Objective-oriented Principal Statistician with expertise in developing and applying statistical methods for medical research. Highly skilled in Bayesian statistics, longitudinal data, mixed models, generalized linear models, nonparametric methods, genomics and high-content imaging
- Proven track record of leading statistical strategies for Phase I and II clinical trials in immunology and ophthalmology, driving impactful contributions to drug development and regulatory-facing documentation
- Published **17** (**6** first/co-first author) papers in peer-reviewed journals
- Renowned for exceptional communication, collaboration, and leadership skills
- Recognized for driving innovation, mentoring teams, and delivering end-to-end solutions in data science and biostatistics

EDUCATION

University of North Carolina at Chapel Hill, Chapel Hill, NC

- **Ph.D.** in Biostatistics Advisor: Dr. Joseph G. Ibrahim and Dr. Michael I. Love August 2019
- **M.S.** in Biostatistics Advisor: Dr. Michael G. Hudgens May 2015

Capital University of Economics and Business, Beijing, China

- **B.S.** in Statistics (China National Scholarship and Honor Graduate) June 2013

WORK EXPERIENCE

Genentech, Inc., South San Francisco, CA

Sept 2021 - Present

Principal Statistician (Assoc Director equivalent)

- Lead Statistician for multiple Phase I and Phase II trials in infectious disease, nephrology, geographic atrophy (GA) and intermediate age-related macular degeneration (iAMD), responsible for clinical development plan, trial design, randomization and unblinding, data analysis plan, biomarker strategy, data monitoring committee, database lock, regulatory-facing document authoring
- Define and drive the AI strategy for early clinical development, including the creation of an Agentic AI system for dynamic clinical trial monitoring and decision support in regulated environments, and promoting team adoption of new technologies
- Lead the data science team in assessing the added value of digital devices in chronic cough studies, fostering cross-functional collaboration with Clinical Science, Digital Biomarker, Patient-Reported Outcome teams, and external vendors to deliver impactful insights
- Direct and spearhead the working group in evaluating and pioneering the use of dynamic borrowing of historical control data in retinal diseases, uniting expertise across Biostatistics, Real-World Evidence, and Digital and Imaging Endpoints to drive innovation and enhance decision-making

Clinical Science Rotation (part-time for six months)

- Hands-on work to support both biostatistics and clinical science functions for Vixarelimab phase II study for Idiopathic Pulmonary Fibrosis (IPF) cohort
- Investigated intercurrent event (ICE) handling strategies with different missing value imputation methods for IPF to support the RtQ of statistical analysis plan (SAP) from EMEA
- Practiced medical data review and protocol deviation queries, provided comments for data with questions

Principal Scientist and Group Lead in Research Biostatistics
Senior Scientist and Group Lead in Research Biostatistics

- Lead the development of the statistical analysis methods and workflows of bulk and single-cell amplicon-seq data for validating a CRISPR base editing platform for accelerated drug-resistant variant discovery. The co-first authored manuscript is published on *Cell Reports*
- Co-lead a team across seven functions to build an end-to-end solution using CRISPR system with image readout Optical Pooled Screen (OPS) for target identification. Designed step-by-step roadmaps and timelines, planned day-to-day activities, negotiated resources and budgets, hired and mentored junior data scientists, and prototyped analysis and visualizations
- Implemented a python workflow for fluorescence cell image processing, image quality check, segmentation, in-situ sequencing barcode calling, immunofluorescence and cell paint phenotype features extraction, statistical analysis and visualization for OPS screens, and contributed it to the internal unified OPS software. The co-authored manuscript submitted to *Nature Methods*
- Lead the processing of raw OPS images of colorectal cancer (CRC) cell lines, deep learning models for signal extraction, and statistical analysis of single cell morphological features for target identification and validation

23andMe, Inc., South San Francisco, CA

Sept 2019 - Sept 2021

Scientist I, Computational Biology, Therapeutics

- Apply statistical methods and build computation tools to analyze the 23andMe genetic database and integrate external genomics datasets to identify novel therapeutic targets
- Conducted Loss-of-Function variants analysis and build the interactive html report for target evaluation
- Polygenic Risk Score (PRS) modeling to impute biomarker values for 23andMe genetic database using UK-Biobank data
- Deep learning algorithms for variant-to-gene mapping of GWAS hits integrating eQTL, epigenetic annotations, functional genomics data and annotations
- Provide statistical support to ongoing therapeutic research and development programs including trial design for the phase I study of 23ME-006

UNC Lineberger Comprehensive Cancer Center, Chapel Hill, NC

May 2017 – Aug 2019

Graduate Research Assistant, Biostatistics Core

- Collaborated with oncologists to strategize and/or conduct experiment design, survey sampling, data management and statistical analysis on multiple clinical and genomics projects
- Provided biostatistics consultation to cancer center members every week

Clinical Projects

- Processed data from multiple sources including clinical data from eCRFs, survey data, EMR, patient registries, PRO, clinician reports, experiment data and wearable devices data; familiar with database software like REDCap, OnCore and Qualtrics
- Planned and conducted survival analysis, generalized linear models, mixed effect models and survey sampling methods
- Reviewed study protocols, wrote analysis plans and prepared manuscripts (11 published papers)

Genomics Projects

- Applied Hidden Markov Model (HMM) on ATAC-seq and CHIP-seq data to examine the chromatin status across different cell lines and age groups
- Applied multivariate analysis and linear mixed effects models on the proteomics data of kinases expression over time to explore the drug targets
- Performed Gene Ontology (GO) and network analysis using DAVID software

- Evaluated a quantification pipeline of label-free mass spectrometry-based proteomics built on OpenMS

Genentech, Inc., South San Francisco, CA

June 2018 - August 2018

Intern, Cancer Immunology

- Analyzed RNA-Seq expression data sets of two single-cell experiments to discover potential targets for combinatorial immune-therapy and to understand the biological functions of a transcription factor
- Integrated an analysis pipeline of single-cell RNA-Seq data sets from data quality control, normalization, controlling nuisance covariates, clustering to gene and gene sets differential expression analysis
- Collaborated with immunologists on analysis plan and results delivery with biological evidences, resulting in a presentation to a team of immunologists

RESEARCH EXPERIENCE

Statistical Methods for Sequencing Count Data and Integrative Functional Genomics (Dissertation)

Oct 2016 – Aug 2019

- Proposed an empirical Bayes method with adaptive t priors to the coefficients in Negative Binomial GLMs to stabilize the LFC estimation, and various numerical techniques for approximation of the posterior distribution
- Proposed nonparametric rank tests with Gibbs or bootstrap samples to test differential expression of RNA-seq with quantification uncertainty and using permutation to build the null distribution
- Proposed Bayesian hierarchical models to determine the colocalization of GWAS and eQTL signals and further estimate the causal effect of genes to complex traits

Causal Effects of Drugs / Drug-Drug Interactions on Adverse Event using EMR Data (Advisor: Dr. Donglin Zeng)

Sept 2014 - May 2017

- Causal inference of matched case-control design with five million samples from EMR to determine the drug and drug-drug interaction (DDI) effects on the adverse event myopathy
- Proposed a conditional causal log-Odds Ratio (OR) definition to characterize individual causal effects, and a control-based propensity score method with spline estimation to estimate this conditional causal log-OR with observational data for better confounders adjustment that provided consistent estimates
- Proposed a nonparametric supreme testing procedure of weighted combination of multiple single effects with logistic regression models to optimize the power in detecting DDI induced adverse events that improved power and controlled type I error in simulation study
- Tackled computational challenges of large-scale health record databases by dimension reduction techniques and meta-analysis methods

SELECT PUBLICATIONS (For a full list of publications, please visit my [google scholar page](#))

1. Dorighi K*, **Zhu A***, Fortin JP, Lo JH, Sudhamsu J, Durinck S, Callow M, Foster S, Haley B, (2024). Accelerated drug resistant variant discovery with an enhanced, scalable mutagenic base editor platform. *Cell Reports*, 43(6), e114314.
2. **Zhu A***, Matoba N*, Wilson E, Tapia AL, Li Y, Ibrahim JG, Stein JL and Love MI, (2021). MRLocus: identifying causal genes mediating a trait through Bayesian estimation of allelic heterogeneity. *PLoS genetics* 17 (4), e1009455.
3. **Zhu A**, Zeng D, Shen L, Ning X, Li L, and Zhang P, (2020). A super-combo-drug test to detect adverse drug events and drug interactions from electronic health records in the era of polypharmacy. *Statistics in Medicine*, 39(10), 1458-1472.

4. **Zhu A**, Ibrahim JG and Love MI, (2019). Heavy-tailed prior distributions for sequence count data: removing the noise and preserving large differences. *Bioinformatics*. 35(12), 2084-2092.
5. **Zhu A**, Srivastava A, Ibrahim JG, Patro R and Love MI, (2019). Nonparametric expression analysis using inferential replicate counts, *Nucleic Acids Research*, 47(18), e105.
6. **Zhu A**, Zeng D, Zhang P and Li L, (2019). Estimating causal log-odds ratio using the case-control sample and its application in the pharmaco-epidemiology study. *Statistical methods in medical research*, p.0962280217750175.
7. Szeto A, Bucci T, Deal A, **Zhu A**, Ahmad M, Cass A, Sketch M, Kemper R, Zeidner J, Foster M, Muluneh B, and Crona D, (2022). Response to Tyrosine Kinase Inhibitors in Real-World Patients With Chronic Myeloid Leukemia. *Annals of Pharmacotherapy* 56 (7), 753-763.
8. Ehlers M, Bjurlin M, Gore J, Pruthi R, Narang G, Tan R, Nielsen M, **Zhu A**, Deal A, and Smith A, (2021). A national cross-sectional survey of financial toxicity among bladder cancer patients. *Urologic Oncology: Seminars and Original Investigations* 39 (1), 76. e1-76. e7.
9. Malpica Castillo LE, Palmer S, **Zhu A**, Deal AM, Chen SL, and Moll S, (2020). Incidence and time course of neutropenia in patients treated with rituximab-based therapy for non-malignant immune-mediated hematologic diseases. *Am J Hematol*. 2020;95(5):E117-E120. doi:10.1002/ajh.25751.

R PACKAGES

1. **Zhu A**, Ibrahim JG and Love MI (2017). **apeglm**: Approximate Posterior Estimation for GLM Coefficients. R package with the *Bioconductor* project.
2. **Zhu A**, Srivastava A, Ibrahim J, Patro R, Love M, (2019). **fishpond**: differential transcript and gene expression with inferential replicates. R package with the *Bioconductor* project.
3. **Zhu A**, Matoba N, Stein J, Love M (2020). **MRLocus**: Mendelian Randomization Analysis per Locus. R package on *github*.

SELECT PRESENTATIONS AND LECTURES

1. **Oral Presentation** “Identifying new targets for colorectal cancer using optical pooled screens”, gRED Group Leader Offsite, Napa CA, June 2023
2. **Invited Talk** “Understanding gene-to-trait effect with GWAS and eQTL summary statistics with Bayesian hierarchical modeling”, ICSA2021, online, September 2021
3. **Oral Presentation** “Nonparametric expression analysis using inferential replicate counts,” JSM2019, Denver CO, July 2019
4. **Oral Presentation** “Application of *t* priors to sequence count data: removing the noise and preserving large differences,” ENAR2018, Atlanta GA, March 2018

TECHNICAL SKILLS

- R for statistical analysis and software development, SAS for data management, processing and analysis and SQL for data query
- Python for fluorescence microscopy cell image processing, data analysis, visualization and deep learning models
- SAS Certified Base Programmer and SAS Certified Advanced Programmer
- Extensive experience with working under the UNIX environment and bash script writing

PROFESSIONAL TRAININGS AND SERVICES

- **Deep Learning Specialization Certificate**
DeepLearning.AI, Coursera, May 2020
- **Meetup Co-lead**, Bay area Biotech-pharma Statistical Workshop (BBSW)
San Francisco Bay Area, June 2020 - Present